

Phase 17 – Production-Style CI/CD & Observability

Containerized Flask Application on AWS

1. Development (Code)


GitHub Repository
project-revision

 app.py
 Dockerfile
 requirements.txt
 .github/workflows/

2. CI/CD (GitHub Actions)


GitHub Actions

Run Tests
↓
Build Docker Image
↓
Push to ECR

3. Container Registry (Amazon ECR)



Amazon ECR
phase17-flask

Docker Image
 Tag:
54f83099270b...

Stored in ECR
(versioned by Git SHA)

4. AWS Infrastructure (Production)

 AWS Cloud (us-east-1)

 **VPC**
(Custom VPC)


Launch Template
phase13-flask-lt
(v5)


Auto Scaling Group
phase13-flask-asg
(Min: 2, Desired: 2, Max: 4)

 us-east-1a


EC2 Instance
Flask (Docker)

 us-east-1b


EC2 Instance
Flask (Docker)


Application Load Balancer
phase17-flask-alb
(Internet-facing)

Target Group
phase13-flask-tg
(HTTP : 8000)
Health Check: /health


Users
(Internet)

▼ 5. Monitoring & Observability (Amazon CloudWatch)



CloudWatch Metrics

- > ALB Request Count
- > ALB Target Health
- > ALB Response Time
- > ALB 5XX Errors
- > ASG CPU Utilization
- > ASG Capacity

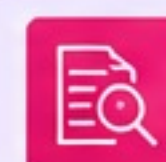


CloudWatch Dashboard Phase17-Production



CloudWatch Alarms

-  **Phase17-ALB-Unhealthy-Targets**
(Alert on unhealthy targets)
-  **Phase17-ALB-Target-5XX**
(Alert on 5XX errors)
-  **ASG Target Tracking Alarm**
(Scale based on CPU utilization)



CloudWatch Logs /phase17/flask

ip-172-31-88-140.ec2.internal
ip-172-31-3-78.ec2.internal
ip-172-31-11-112.ec2.internal
ip-172-31-84-153.ec2.internal
...

▼ Details

Load balancer type Application	Status ✔ Active	VPC vpc-0e5b9a23d60466a0e ↗	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-04f6cc63e6e3c38a8 ↗ us-east-1b (use1-az2) subnet-011d2dca563075474 ↗ us-east-1f (use1-az5) subnet-01205bdc921170480 ↗ us-east-1a (use1-az1)	Date created October 4, 2026, 18:07 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:207224908189:loadbalancer/app/phase17-flask-alb/e7ed5077b258edce		DNS name Info phase17-flask-alb-887218244.us-east-1.elb.amazonaws.com (A Record)	

- Listeners and rules
- Network mapping
- Resource map
- Security
- Monitoring
- Integrations
- Attributes
- Capacity
- Tags

Listeners and rules (1) [Info](#)

Manage rules ▾

Manage listener ▾

Add listener

A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.

Filter listeners

< 1 >

☐	Protocol:Port ▾	Default action ▾	Rules ▾	ARN ▾	Security policy ▾	Default SSL/TLS cer... ▾	mTLS ▾	Trust s
☐	HTTP:80	• Forward to target group phase13-flask-tg ↗: 1 (100%) Target group stickiness: Off	1 rule	A...	Not applicable	Not applicable	Not applicable	Not ap

Not Secure — phase17-flask-alb-887218244.us-east-1.elb.amazonaws.com

DEVOPS CONTINUED | Phase 17 - Production CD · Workflow... | Target group details | EC2 | us-east-1 | Amazon Web Services Sign-in | phase17-flask-alb-887218244.us-e...

```
{ "status": "healthy" }
```

Phase17-Production ☆

1h 3h 12h 1d 3d 1w Custom Local timezone Actions Save

Phase 17 — Production Observability

Architecture: GitHub Actions → ECR → Launch Template → ASG → ALB → Flask

Production image: 54f83099270bc455d4a0a5247e137c1d0a5b1ada



▼ Log group details

CloudWatch console feature

ARN

arn:aws:logs:us-east-1:207224908189:log-group:/phase17/flask:*

Creation time

10 hours ago

Retention

2 weeks

Stored bytes

-

Metric filters

0

Subscription filters

0

Contributor Insights rules

-

KMS key ID

-

Deletion protection

Off

Data protection

-

Sensitive data count

-

Custom field indexes

[Configure](#)

Transformer

[Configure](#)

Anomaly detection

[Configure](#)

Bearer token authentication

Off

Syslog ingestion

[Info](#)

[Configure](#)

Log streams (5)

By default, we only load the most recent log streams.

☐ Exact match ☐ Show expired [Info](#)

Log stream

ip-172-31-88-40.ec2.internal

ip-172-31-3-78.ec2.internal

ip-172-31-88-140.ec2.internal

ip-172-31-11-112.ec2.internal

ip-172-31-84-153.ec2.internal

Last event time

2026-10-05 04:53:07 (UTC+05:30)

2026-10-05 04:48:19 (UTC+05:30)

2026-10-05 04:48:17 (UTC+05:30)

2026-10-04 19:40:43 (UTC+05:30)

2026-10-04 19:40:40 (UTC+05:30)

Refresh Delete Create log stream Search all log streams

< 1 >

Alarms (5)

Filter alarms

Hide Auto Scaling alarms

< 1 >

TargetTracking-phase13-flask-asg-AlarmLow-e1fa0b6c-3519-4d11-9b97-0a5d3601fe1f

Metric alarm

In alarm

Phase17-ALB-Unhealthy-Targets

Metric alarm

OK

TargetTracking-phase13-flask-asg-AlarmHigh-ed196f85-4395-4370-a2af-33a0ff7fd131

Metric alarm

OK

Phase17-ALB-Target-SXX

Metric alarm

OK

Mem_Alarm

Metric alarm

OK



Details

Tags

Mute rules

Actions

History

Parent alarms

CloudWatch console feature

Details

Name

Phase17-ALB-Unhealthy-Targets

Type

Metric alarm

Description

Alert when the Phase 17 ALB has one or more unhealthy targets

State

OK

Threshold

UnHealthyHostCount > 0 for 2 datapoints within 2 minutes

Actions

No actions

Last state update

2026-10-05 04:58:51 (Local)

Namespace

AWS/ApplicationELB

Metric name

UnHealthyHostCount

TargetGroup

targetgroup/phase13-flask-tg/1f9e3e1e7b324945

LoadBalancer

app/phase17-flask-alb/e7ed5077b258edce

Statistic

Maximum

Period

1 minute

Datapoints to alarm

2 out of 2

Missing data treatment

Treat missing data as good (not breaching threshold)

Percentiles with low samples

evaluate

ARN

arn:aws:cloudwatch:us-east-1:207224908189:alarm:Phase17-ALB-Unhealthy-Targets

View EventBridge rule